# 802.11be vs 802.11n: P-EDCA Tail-Latency Comparison

CWds × QSRC × PSRC parameter sweep | CBR / Poisson / MMPP / OnOff traffic | P-EDCA STA perspective

## Simulation setup

| | |
|---|---|
| **Topology** | 30 STAs → 1 AP, random 1–5 m disc; 5 GHz ch36 / 20 MHz |
| **Traffic** | Uplink AC_VO UDP, 1 Mbps per STA (1000 B); CBR / Poisson / MMPP / OnOff |
| **PHY** | 11n = HtMcs7 (65 Mbps) vs 11be = EhtMcs7 (73.1 Mbps @ 3.2 µs GI) |
| **Control** | OFDM 6 Mbps for both; A-MPDU on; VO TXOP limit 2.08 ms |
| **Fairness** | The two programs differ by exactly 2 lines (SetStandard / DataMode) |
| **Sweep** | nPedca ∈ {5, 15, 30}; CWds{0,1} × QSRC{0–5} × PSRC{1–3} = 36 combos |
| **Statistics** | 10 seeds × 10 s per point; plus an EDCA-only baseline (nPedca = 0) |

Data integrity: 4 traffic × 290 delay histograms are independent runs (11n / 11be md5 all differ)

**P99 −63% ~ −45%**

Tail latency at full P-EDCA load (30/30), best params

Consistent across 4 traffic; 11be 6.3–11.5 ms vs 11n 17–21 ms

**Loss 7–9% → 2–4%**

VO retry-limit loss, EDCA-only baseline

Same params & load; 11n stuck in collision feedback, 11be not

**+35% ~ +54% vs EDCA**

P-EDCA still pays off at full penetration on 11be

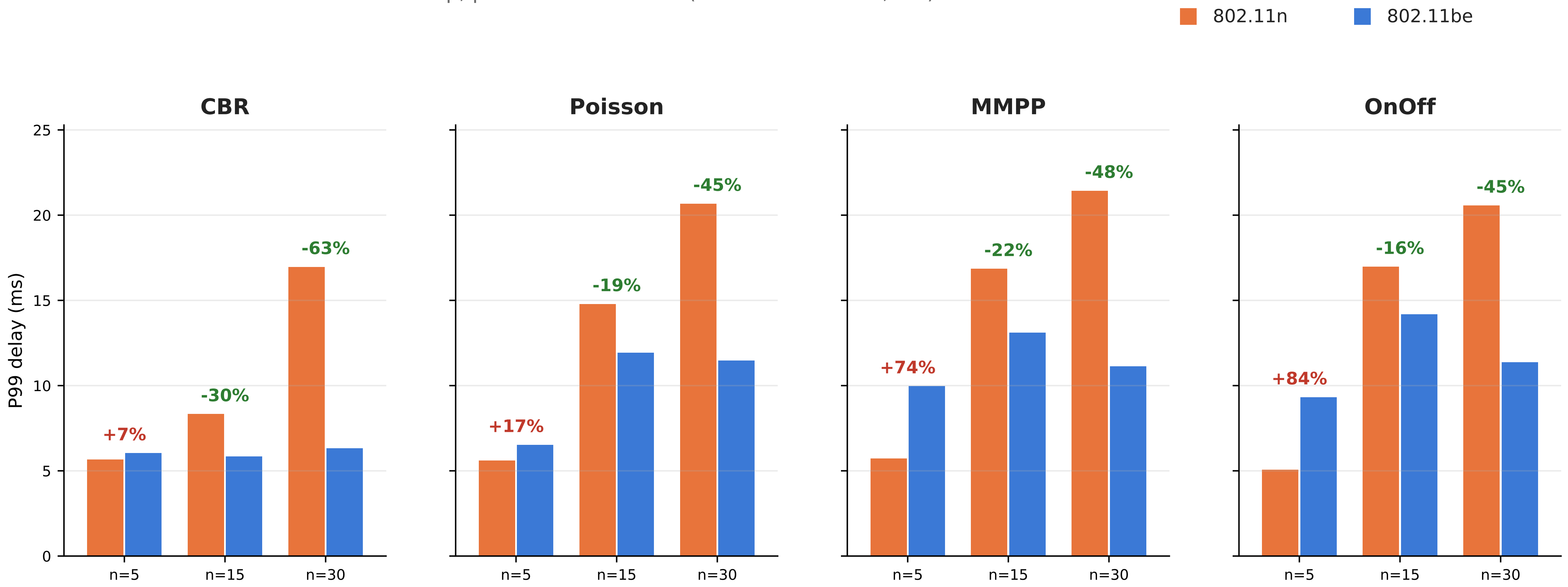
Only +3% to +13% on 11n — P-EDCA value grows with the PHY generation

### One-line takeaway

11be's edge is NOT faster per-frame airtime (access delay nearly identical) — it pulls the system off the high-collision operating point: at heavy load the tail roughly halves and the P-EDCA mechanism becomes worthwhile again.

# Core result: P-EDCA P99 latency — 11be's edge grows with load

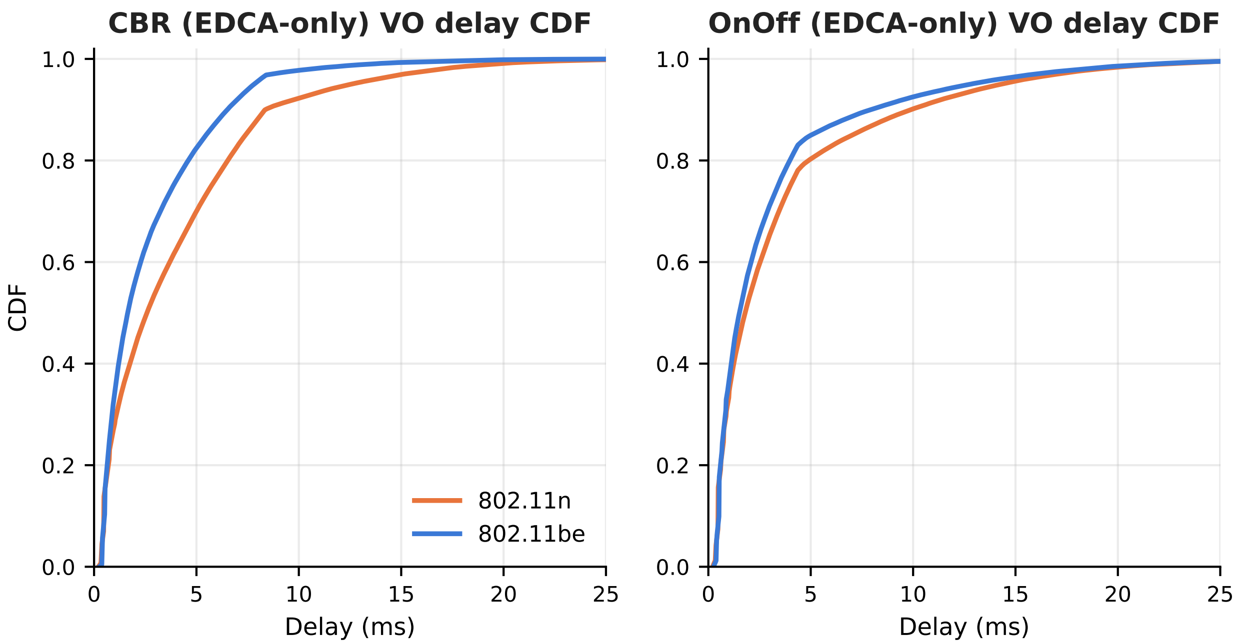
Minimum P99 over the 36-combo sweep, per traffic × nPedca (mean of 10 seeds; ms)

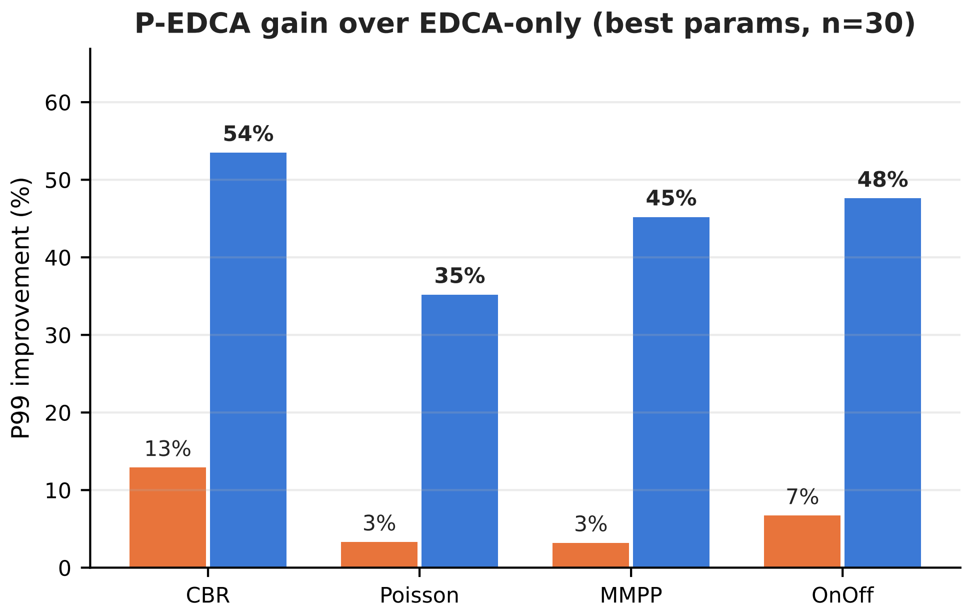



- Light P-EDCA load n=5: 11be shows NO gain, even a penalty (CBR +6%, bursty +74–84%) — few P-EDCA STAs, the priority channel is uncongested and the faster PHY does not help the subset tail.
- Medium n=15: 11be −16% to −29%.  Heavy n=30: 11be −45% to −63% (11n 17–21 ms vs 11be 6.3–11.5 ms).
- The n=5 penalty is consistent across the whole 36-combo distribution (medians too), not a min-artifact — it is the low-contention regime where the P-EDCA subset tail is arrival-driven and noisy.
- Best combos: both PHYs prefer aggressive PSRC=3 and small QSRC (0–2); 11n's slow large-CW retries under load still cannot recover → saturated at 17–21 ms.

# Mechanism: queueing & collisions, not frame speed

The whole delay distribution shifts left. EDCA-only baseline (no P-EDCA, 30 STAs) isolates the two PHY generations.




- P10: 11be +7~+11% (longer EHT preamble on one uncontended access); P50: -36~-16%; P99: −2~−30% — same floor, 11be pulls ahead from the median toward the tail.
- PHY gain shrinks with burstiness: EDCA-only P99 gain CBR −30% → Poisson −17% → MMPP −8% → OnOff −2% (burst-scale queueing is arrival-driven).
- P-EDCA view: on loaded 11n the DS-CTS overhead barely pays (+3~+13%); on 11be the same mechanism still has headroom and yields +35~+54%.

# Root cause: different fixed points of the collision-queueing feedback loop

Per-frame airtime is nearly identical (EHT +12% payload rate, longer preamble) — the cause is the operating point, not raw speed

## Causal chain (11n stuck at the high-collision fixed point; 11be at the low one)

EhtMcs7 service margin: ~12% more payload per 2.08 ms TXOP; retry rounds slightly shorter

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Service headroom up, queue sojourn down, fewer simultaneously backlogged STAs

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VO CWmin=3 (4 slots): collision rate very sensitive to contender count → collisions drop

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Shorter retry chains; retry-limit loss 7–9% → 2–4% (measured, EDCA-only)

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Feedback settles at a low-collision fixed point: distribution shifts, P99 −2% to −30%

## Supporting evidence (EDCA-only, 30 STAs; 11n → 11be)

| Traffic | VO collision loss | Queue delay (µs) | Access delay (µs) | Channel idle |
|---|---|---|---|---|
| CBR | 9.1% → 1.9% | 3636 → 2428 | 312 → 297 | 25.1% → 24.9% |
| Poisson | 9.1% → 2.9% | 3904 → 2720 | 366 → 364 | 23.4% → 22.9% |
| MMPP | 8.9% → 4.1% | 3904 → 3051 | 393 → 409 | 22.8% → 22.1% |
| OnOff | 7.0% → 3.9% | 3174 → 2669 | 421 → 441 | 22.4% → 22.2% |

Reading: queue delay is ~90% of MAC delay and drops 16–33%; access delay barely differs (<5%, even slightly higher for 11be on bursty — longer PPDU); idle differs <1 pp; carried throughput equal.

## Open items (suggested follow-ups)

Whether the ~12% service margin alone explains the fixed-point gap, or ns-3's HT vs EHT frame-exchange / receive paths (post-collision EIFS behavior; this repo's v6–v6.3 changes) also contribute: (1) count PhyRxDrop / collision events per standard; (2) rate-sensitivity run with 11be GI = 0.8 µs (86 Mbps) to separate rate effects from code-path effects.

## Why each pattern appears

### ▪ No gain at light P-EDCA load

Priority channel uncongested; P-EDCA subset tail is arrival-driven, so the faster PHY does not help — and for bursty traffic 11be even trails.

### ▪ Smaller gain when bursty

Burst peaks far exceed the service rate; queueing is arrival-driven, the margin only shortens the drain time.

### ▪ Largest gap at n=30

DS-CTS adds one 6 Mbps exchange per access; 11n adds it past the knee (blow-up); 11be absorbs it.